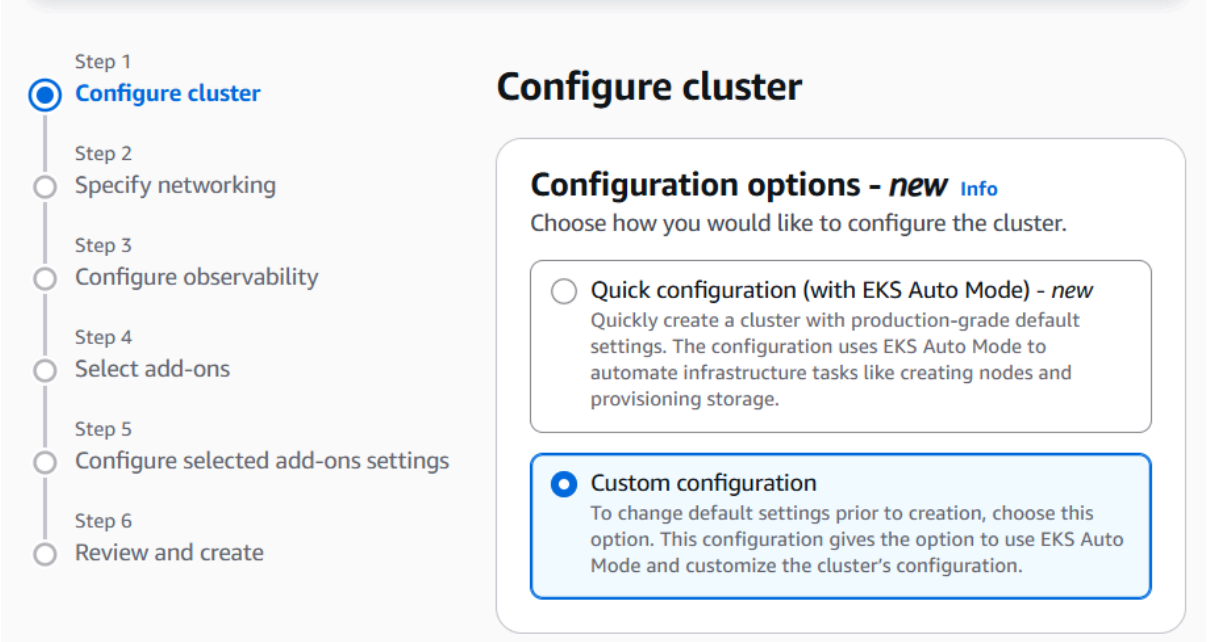
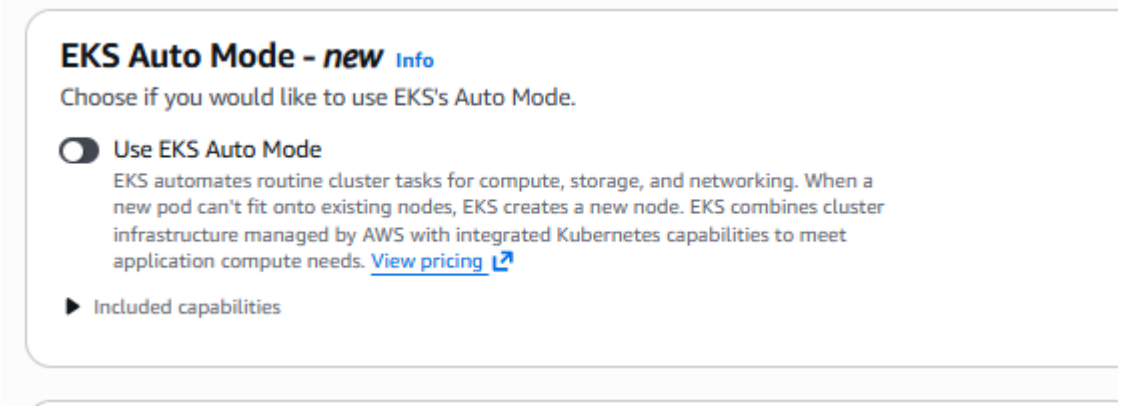
DEVOPS - 11-03-2026

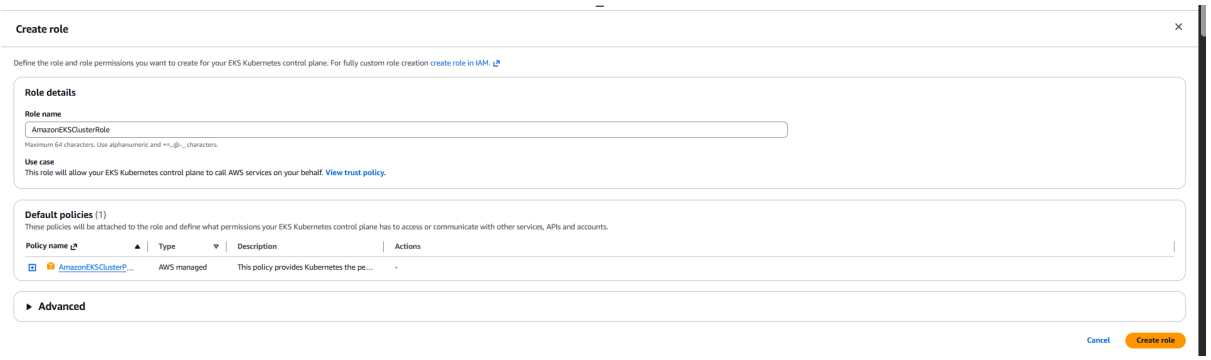
EKS

Create Cluster



STEP 1:





Amazon Elastic Kubernetes Service > Create EKS cluster

Role AmazonEKSClusterRole has been successfully created

Notifications 0 0 1 1 0

Step 2
Specify networking

Step 3
Configure observability

Step 4
Select add-ons

Step 5
Configure selected add-ons settings

Step 6
Review and create

Configuration options - *new* Info

Choose how you would like to configure the cluster.

Quick configuration (with EKS Auto Mode) - *new*
Quickly create a cluster with production-grade default settings. The configuration uses EKS Auto Mode to automate infrastructure tasks like creating nodes and provisioning storage.

Custom configuration
To change default settings prior to creation, choose this option. This configuration gives the option to use EKS Auto Mode and customize the cluster's configuration.

EKS Auto Mode - *new* Info

Choose if you would like to use EKS's Auto Mode.

Use EKS Auto Mode
EKS automates routine cluster tasks for compute, storage, and networking. When a new pod can't fit onto existing nodes, EKS creates a new node. EKS combines cluster infrastructure managed by AWS with integrated Kubernetes capabilities to meet application compute needs. View pricing

Included capabilities

Cluster configuration Info

Name
Enter a unique name for this cluster. This property cannot be changed after the cluster is created.

testeks

The cluster name should begin with letter or digit and can have any of the following characters: the set of Unicode letters, digits, hyphens and underscores. Maximum length of 100.

Cluster IAM role | Info
Select the Cluster IAM role to allow the Kubernetes control plane to manage AWS resources on your behalf. This cannot be changed after the cluster is created. To create a new custom role, follow the instructions in the Amazon EKS User Guide.

AmazonEKSClusterRole

Create recommended role

Kubernetes version settings

Kubernetes version | Info
Select Kubernetes version for this cluster.

1.35

Upgrade policy | Info
Choose one of the following options. You can switch the setting later while the standard support period is in effect.

Standard support
This option supports the Kubernetes version for 14 months after the release date. There is no additional cost. When standard support ends, your cluster will be auto upgraded to the next version.

Extended support
This option supports the Kubernetes version for 26 months after the release date. The extended support period has an additional hourly cost that begins after the standard support period ends. When extended support ends, your cluster will be auto upgraded to the next version.

Control plane scaling tier - *new* Info

Select a scaling tier for your control plane. While Standard mode scales dynamically, higher tiers pre-provision your environment with fixed, high-performance resources. This eliminates scaling latency and provides the sustained throughput required for demanding workloads. View tier pricing

☐ Use a scaling tier
For predictable and high performance from your cluster's control plane, choose a scaling tier.

Auto Mode Compute - *new* Info

Configure node management for your EKS cluster. EKS offers four compute options: EKS Auto Mode, EC2 Managed Node Groups, Fargate, and hybrid nodes. Node groups, Fargate profiles, and hybrid nodes are configured after cluster creation. You can also create self-managed nodes.

Compute configuration
If EKS Auto Mode is not managing compute resources, you need to create compute resources once the cluster is ready. We recommend creating a node group after cluster creation.

View documentation

Cluster access Info

Control how IAM principals can access this cluster.

Bootstrap cluster administrator access | Info
Choose whether the IAM principal creating the cluster has Kubernetes cluster administrator access.

- (●) Allow cluster administrator access
  Allow cluster administrator access for your IAM principal.
- ( ) Disallow cluster administrator access
  Disallow cluster administrator access for your IAM principal.

Cluster authentication mode | Info
Configure which source the cluster will use for authenticated IAM principals.

- (●) EKS API
  The cluster will source authenticated IAM principals only from EKS access entry APIs.
- ( ) EKS API and ConfigMap
  The cluster will source authenticated IAM principals from both EKS access entry APIs and the aws-auth ConfigMap.

Envelope encryption Info

Envelope encryption is applied to all Kubernetes API data.

By default, EKS implements Kubernetes API data envelope encryption using an {AWS} owned key. Alternatively, you can setup your own customer managed key (CMK) and link this key by providing the CMK ARN when configuring your EKS cluster.

◯ Use your own AWS KMS key
After a cluster is created, you can migrate from using an AWS owned key to a customer managed key (CMK), but not vice versa.

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ARC Zonal shift Info

Shift application traffic away from an impaired Availability Zone (AZ) in your EKS cluster. You can change this later.

Enabled

EKS will register your cluster with ARC zonal shift to enable you to use zonal shift to shift application traffic away from an AZ

Disabled

EKS will not register your cluster with ARC zonal shift

Before you start a zonal shift, you need to setup your cluster environment to be resilient to an AZ failure beforehand.

Learn more

Deletion protection

Deletion protection must be turned off to be able to delete a cluster. It can be turned on and off after the cluster is created.

Turn on deletion protection

Deletion protection provides additional security against accidental cluster deletion.

Tags (0) Info

No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Cancel Next

STEP 2:

Step 1
Configure cluster

Step 2
Specify networking

Step 3
Configure observability

Step 4
Select add-ons

Step 5
Configure selected add-ons settings

Step 6
Review and create

Specify networking

Networking Info

IP address family and service IP address range cannot be changed after cluster creation.

VPC | Info

Select a VPC to use for your EKS cluster resources.

vpc-032d74dfeb4fcd867 | Default

Subnets Info

Choose the subnets in your VPC where the control plane may place elastic network interfaces (ENIs) to facilitate communication with your cluster. To create a new subnet, go to the corresponding page in the VPC console.

Select subnets

Clear selected subnets

subnet-0f9bc52c2517a811c
us-east-1c 172.31.16.0/20

subnet-000ddcd63e3337e3d
us-east-1d 172.31.32.0/20

subnet-08f6b7c71cd5bf4a0
us-east-1f 172.31.64.0/20

subnet-0ca08c9499e20ecd6
us-east-1b 172.31.80.0/20

subnet-0d47f687a91c9b321
us-east-1a 172.31.0.0/20

Additional security groups | Info

EKS automatically creates a cluster security group on cluster creation to facilitate communication between worker nodes and control plane. Optionally, choose additional security groups to apply to the EKS-managed Elastic Network Interfaces that are created in your control plane subnets. To create a new security group, go to the corresponding page in the VPC console.

Select security groups

sg-08b342f7558cd55af | launch-wizard-1
launch-wizard-1 created 2026-03-10T05:58:24.250Z

Clear selected security groups

Choose cluster IP address family | Info

Specify the IP address type for pods and services in your cluster.

- (•) IPv4
- ( ) IPv6

Configure Kubernetes service IP address block Info

[ ] Specify the range from which cluster services will receive IP addresses.

Choose cluster IP address family | Info

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- (•) IPv4
- ( ) IPv6

Configure Kubernetes service IP address block Info

[ ] Specify the range from which cluster services will receive IP addresses.

Configure remote networks to enable hybrid nodes Info

EKS Hybrid Nodes enables you to use on-premises and edge infrastructure as nodes in EKS clusters.

[ ] Specify the CIDR blocks for your on-premises environments that you will use for hybrid nodes.

Cluster endpoint access Info

Configure access to the Kubernetes API server endpoint.

- (•) **Public**
  The cluster endpoint is accessible from outside of your VPC. Worker node traffic will leave your VPC to connect to the endpoint.
- ( ) **Public and private**
  The cluster endpoint is accessible from outside of your VPC. Worker node traffic to the endpoint will stay within your VPC.
- ( ) **Private**
  The cluster endpoint is only accessible through your VPC. Worker node traffic to the endpoint will stay within your VPC.

▶ **Advanced settings**

Cancel | Previous | Next

STEP 3:

Step 1
Configure cluster
Step 2
Specify networking
Step 3
Configure observability
Step 4
Select add-ons
Step 5
Configure selected add-ons settings
Step 6
Review and create

Configure observability

▸ About observability

Metrics

Prometheus | Info

☐ Send Prometheus metrics to Amazon Managed Service for Prometheus
Monitor your application and infrastructure metrics with Amazon Managed Service for Prometheus. These metrics include system health and performance data.

ⓘ Agentless Prometheus metrics collection requires the cluster API server to be available privately. To make the following toggle available, select either the Public and private option or the Private option for Cluster endpoint access in Specify networking.

CloudWatch | Info

☐ Send application and infrastructure telemetry to Amazon CloudWatch
Installs the Amazon CloudWatch Observability add-on to send application metrics from CloudWatch APM and infrastructure telemetry from CloudWatch Container insights.

▸ Services and telemetry included

Container network observability Info

Network monitoring | Info

☐ Enable network monitoring
Network monitoring enables you to easily troubleshoot by allowing you to visualize traffic in a service map, explore detailed flow tables for granular traffic analysis, monitor network performance metrics in real-time. Learn more

Control plane logs Info

Send audit and diagnostic logs from the Amazon EKS control plane to CloudWatch Logs.

Container network observability Info

Network monitoring | Info

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Control plane logs Info

Send audit and diagnostic logs from the Amazon EKS control plane to CloudWatch Logs.

☐ API server
Logs pertaining to API requests to the cluster.

☐ Audit
Logs pertaining to cluster access via the Kubernetes API.

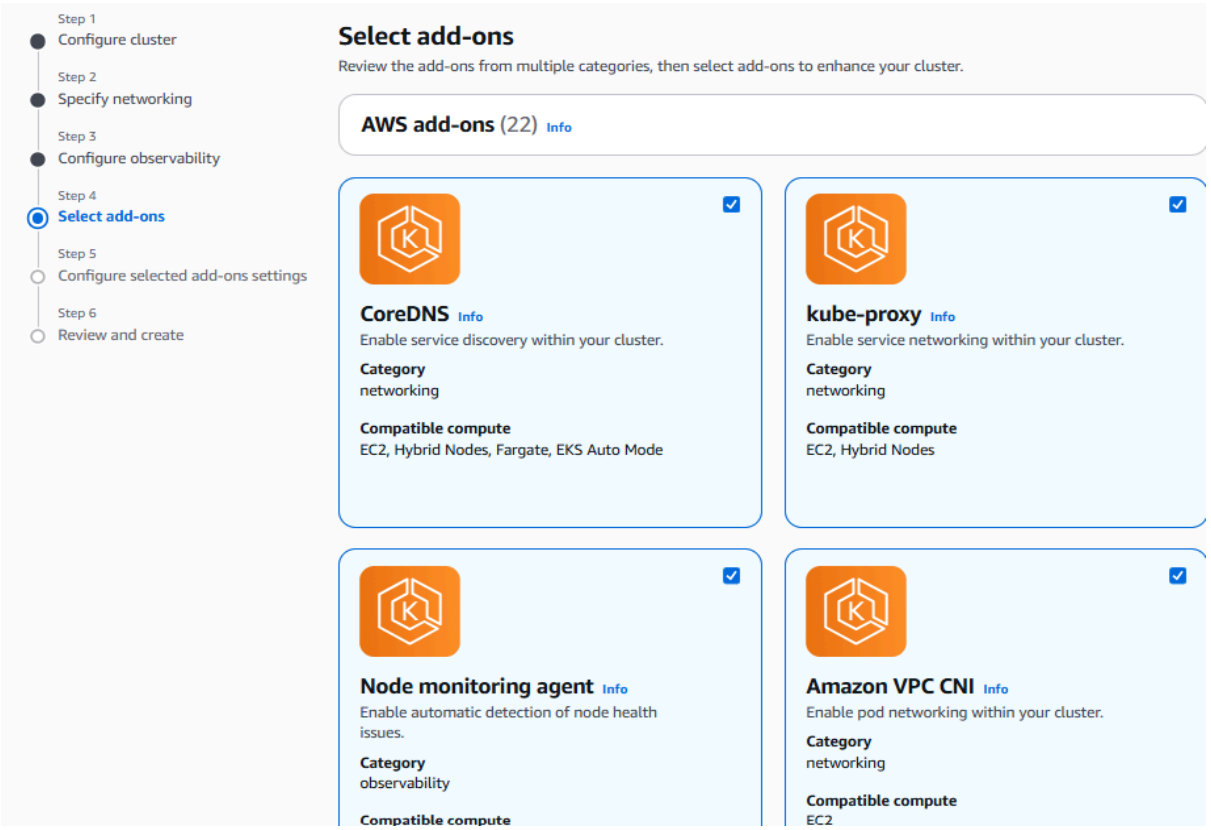
☐ Authenticator
Logs pertaining to authentication requests into the cluster.

☐ Controller manager
Logs pertaining to state of cluster controllers.

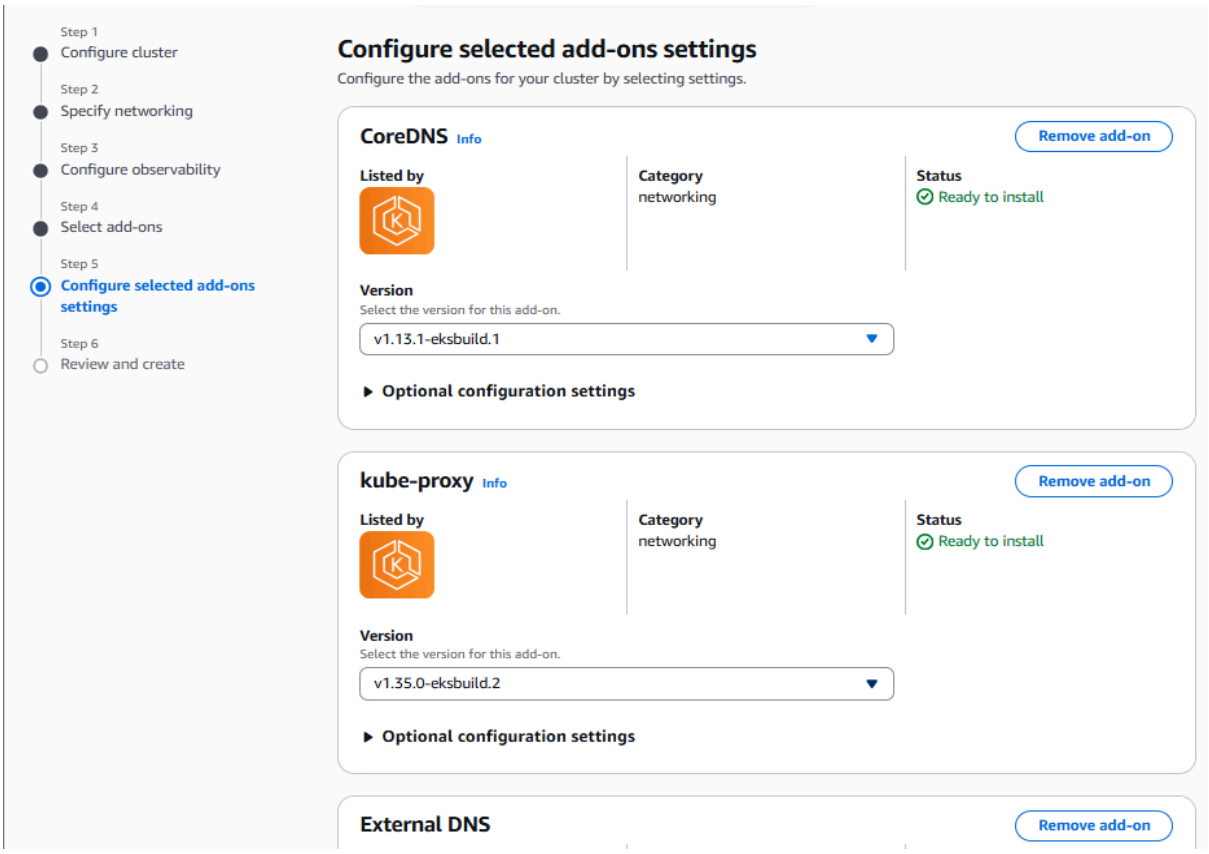
☐ Scheduler
Logs pertaining to scheduling decisions.

Cancel | Previous | Next

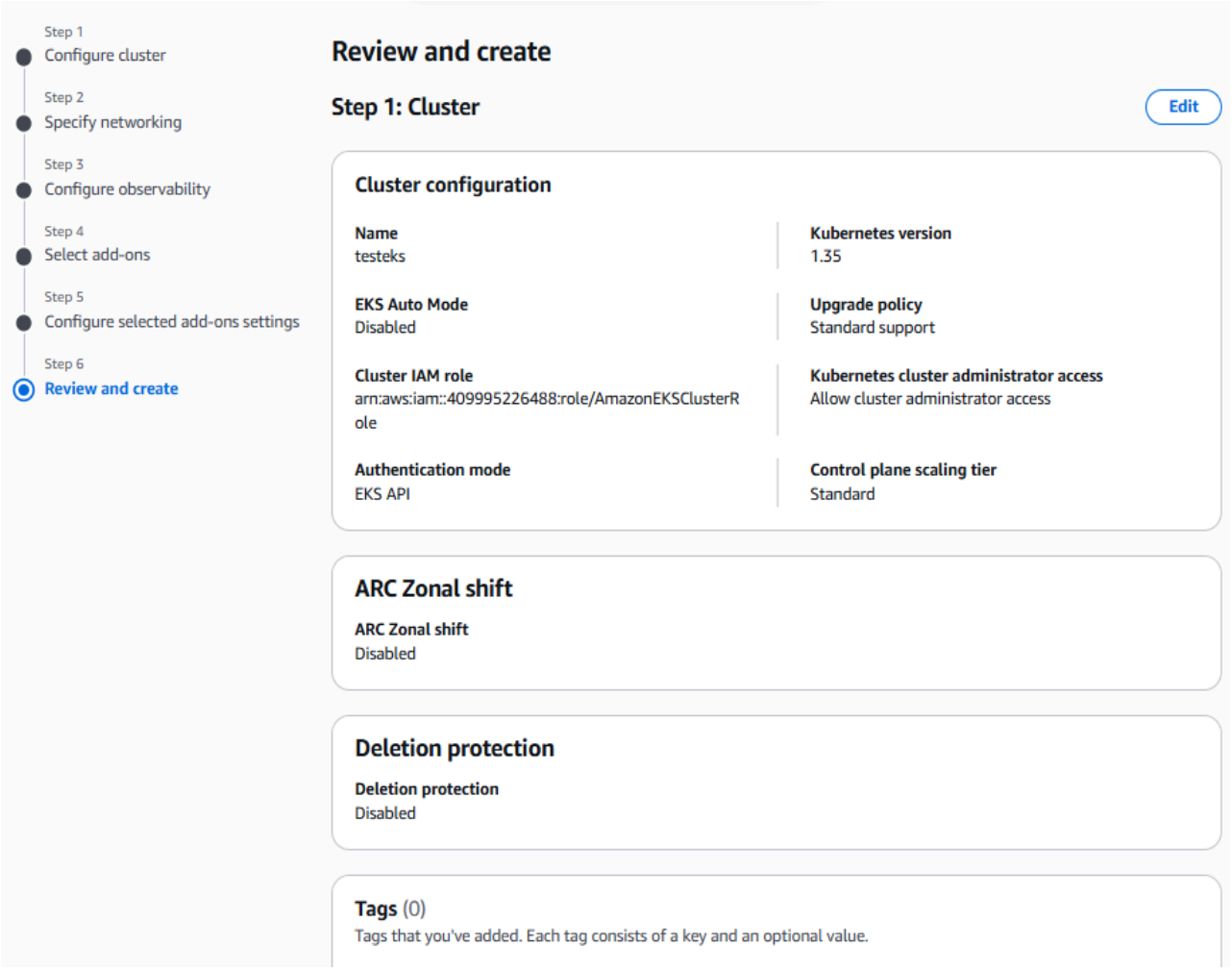
STEP 4:


Step 1
Configure cluster
Step 2
Specify networking
Step 3
Configure observability
Step 4
Select add-ons
Step 5
Configure selected add-ons settings
Step 6
Review and create
Select add-ons
Review the add-ons from multiple categories, then select add-ons to enhance your cluster.
AWS add-ons (22) Info
CoreDNS Info
Enable service discovery within your cluster.
Category
networking
Compatible compute
EC2, Hybrid Nodes, Fargate, EKS Auto Mode
kube-proxy Info
Enable service networking within your cluster.
Category
networking
Compatible compute
EC2, Hybrid Nodes
Node monitoring agent Info
Enable automatic detection of node health issues.
Category
observability
Compatible compute
Amazon VPC CNI Info
Enable pod networking within your cluster.
Category
networking
Compatible compute
EC2

STEP 5:


Step 1
Configure cluster
Step 2
Specify networking
Step 3
Configure observability
Step 4
Select add-ons
Step 5
Configure selected add-ons settings
Step 6
Review and create
Configure selected add-ons settings
Configure the add-ons for your cluster by selecting settings.
CoreDNS Info
Remove add-on
Listed by
Category
networking
Status
Ready to install
Version
Select the version for this add-on.
v1.13.1-eksbuild.1
Optional configuration settings
kube-proxy Info
Remove add-on
Listed by
Category
networking
Status
Ready to install
Version
Select the version for this add-on.
v1.35.0-eksbuild.2
Optional configuration settings
External DNS
Remove add-on

STEP 6:


Step 1
Configure cluster
Step 2
Specify networking
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Configure observability
Step 4
Select add-ons
Step 5
Configure selected add-ons settings
Step 6
Review and create
Review and create
Step 1: Cluster
Edit
Cluster configuration
Name
testeks
Kubernetes version
1.35
EKS Auto Mode
Disabled
Upgrade policy
Standard support
Cluster IAM role
arn:aws:iam::409995226488:role/AmazonEKSClusterRole
Kubernetes cluster administrator access
Allow cluster administrator access
Authentication mode
EKS API
Control plane scaling tier
Standard
ARC Zonal shift
ARC Zonal shift
Disabled
Deletion protection
Deletion protection
Disabled
Tags (0)
Tags that you've added. Each tag consists of a key and an optional value.

Step 5: Versions

Edit

Selected add-ons version (7)

Find add-on versions

1

Add-on name	Version
coredns	v1.13.1-eksbuild.1
eks-node-monitoring-agent	v1.5.2-eksbuild.1
eks-pod-identity-agent	v1.3.10-eksbuild.2
external-dns	v0.20.0-eksbuild.3
kube-proxy	v1.35.0-eksbuild.2
metrics-server	v0.8.1-eksbuild.2
vpc-cni	v1.21.1-eksbuild.1

EKS Pod Identity (2)

1

Add-on name	IAM role	Service account
external-dns	Not set	external-dns
vpc-cni	Not set	aws-node

Cancel Previous Create

Cluster is created

Amazon Elastic Kubernetes Service

Dashboard
Clusters
Settings
Dashboard settings
Console settings
Amazon EKS Anywhere
Enterprise Subscriptions
Related services
Amazon ECR
AWS Batch

Documentation
EKS workshops New

Next step: Provision compute capacity for your cluster by adding a Managed node group or creating a Fargate profile.

Notifications 0 0 0 3 0

testeks

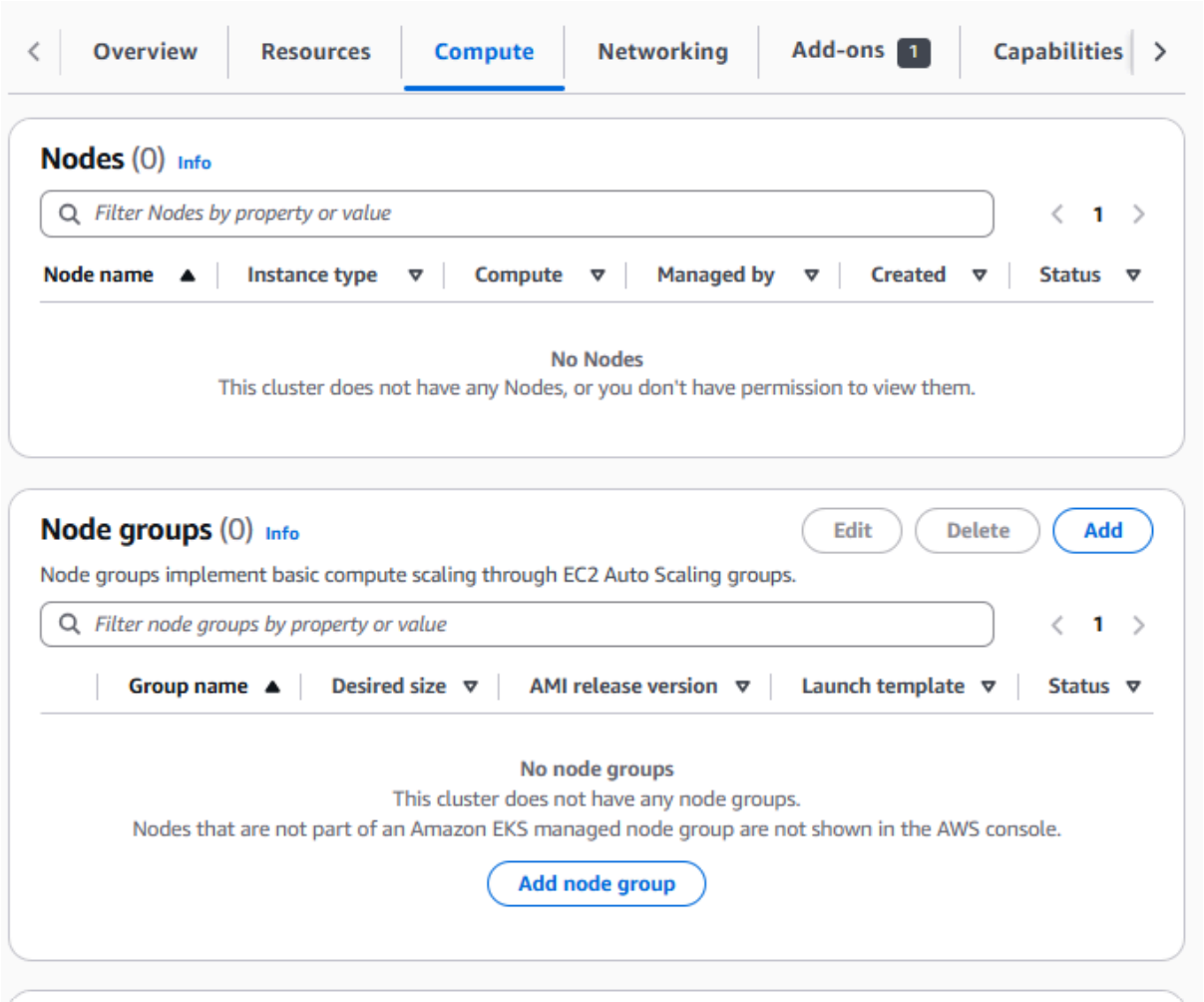
Delete cluster Monitor cluster

Cluster info Info

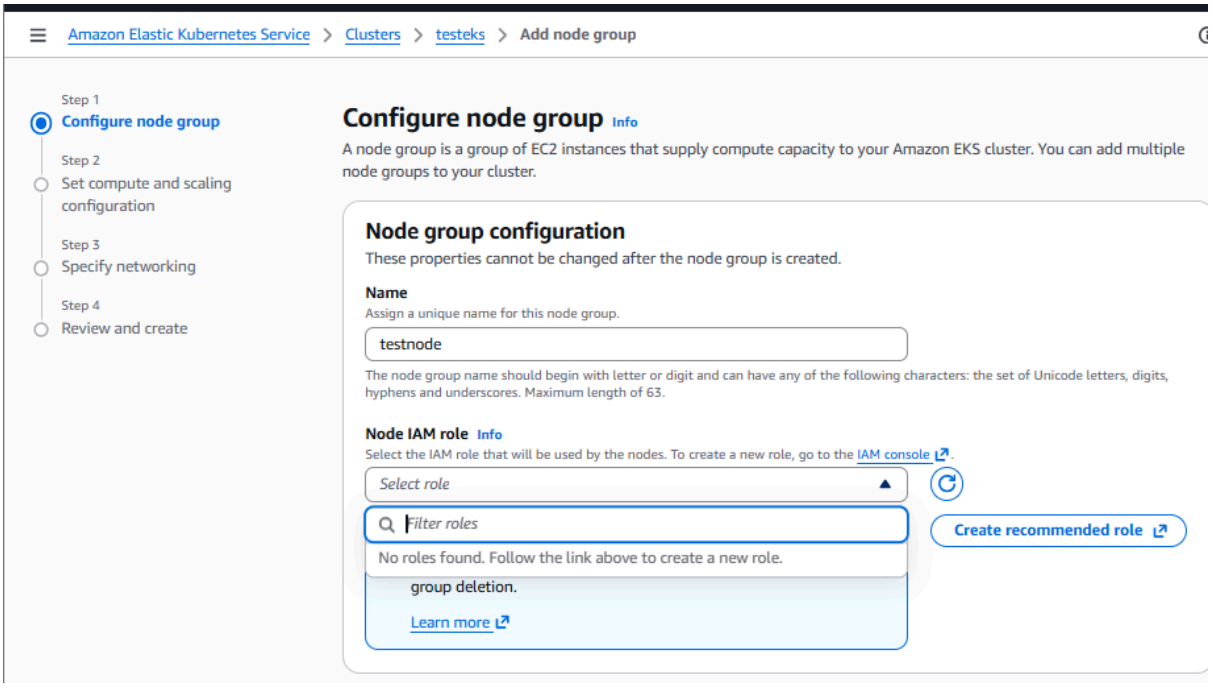
Status	Kubernetes version	Support period	Provider
Active	Info 1.35	Standard support until March 27, 2027	EKS

Cluster health	Upgrade insights	Node health issues	Capability issues
0	5	0	0

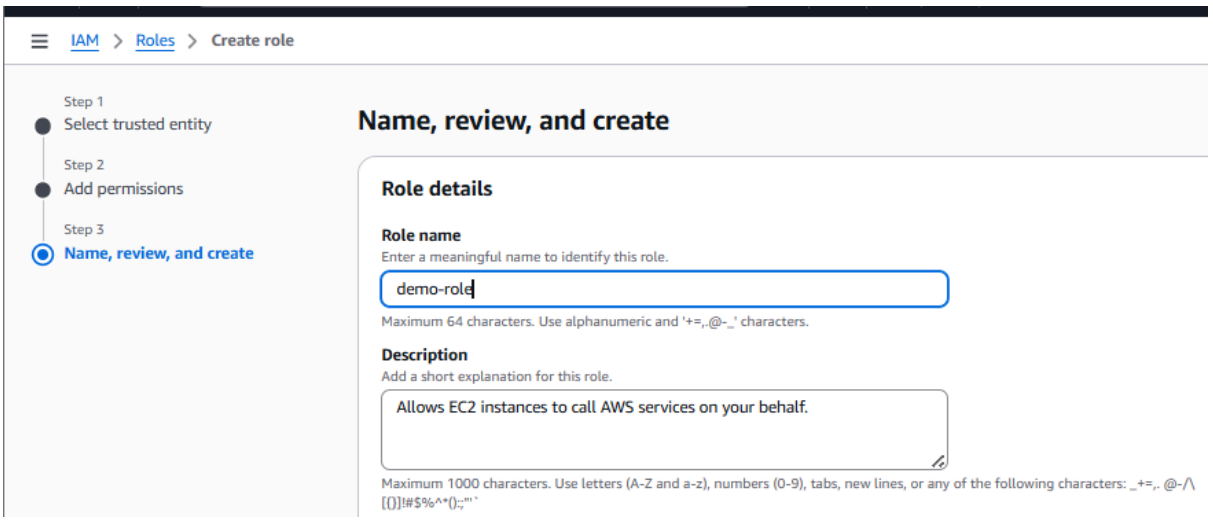
Overview | Resources | Compute | Networking | Add-ons | Capabilities



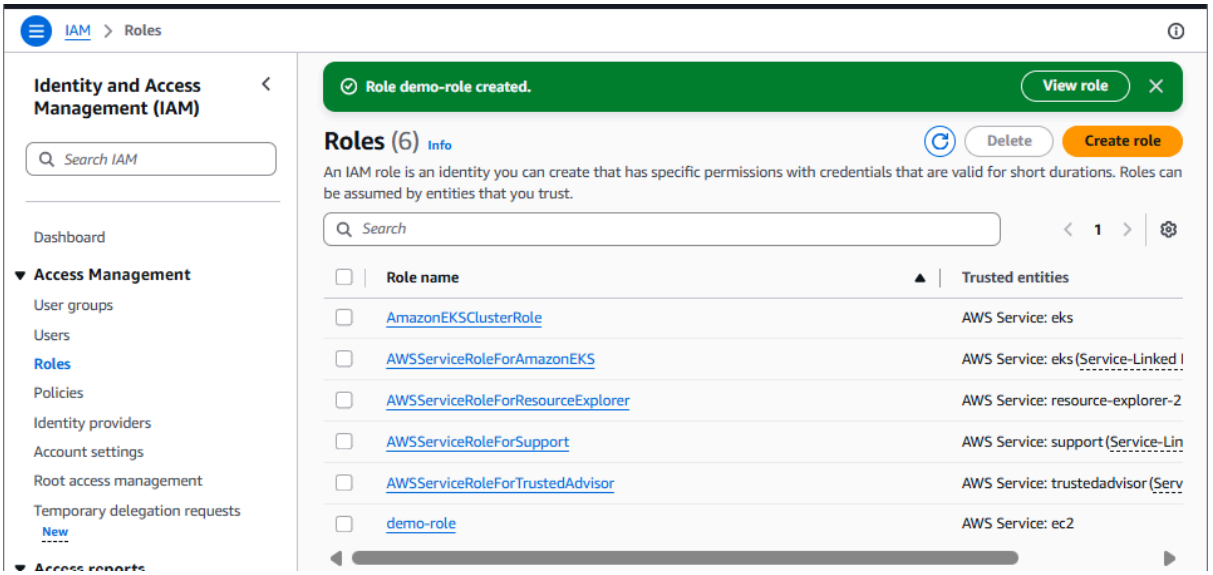
STEP 1:



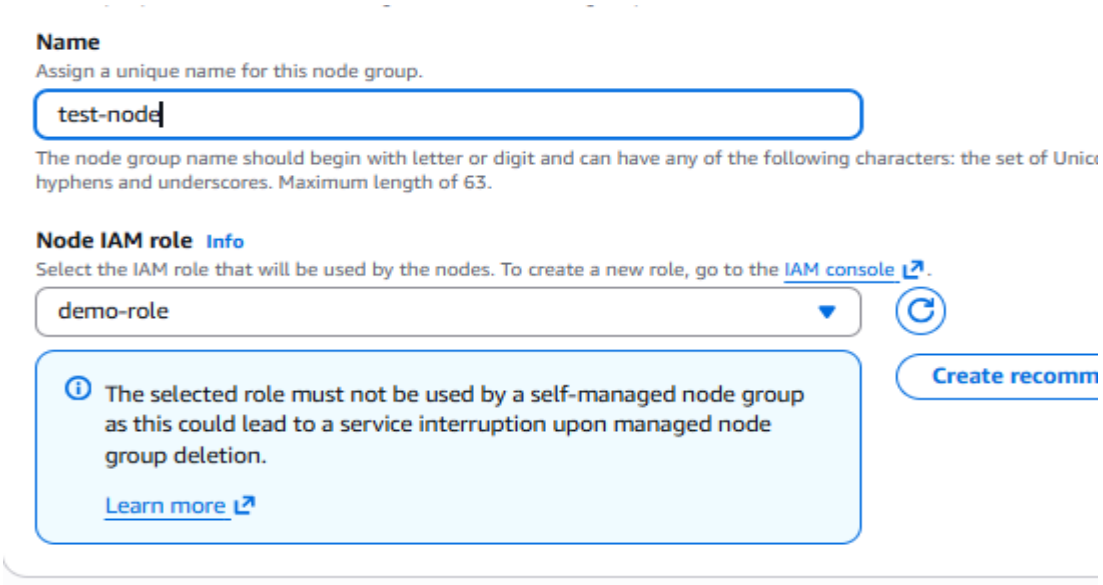
Step 1 & 2 => default data => click next



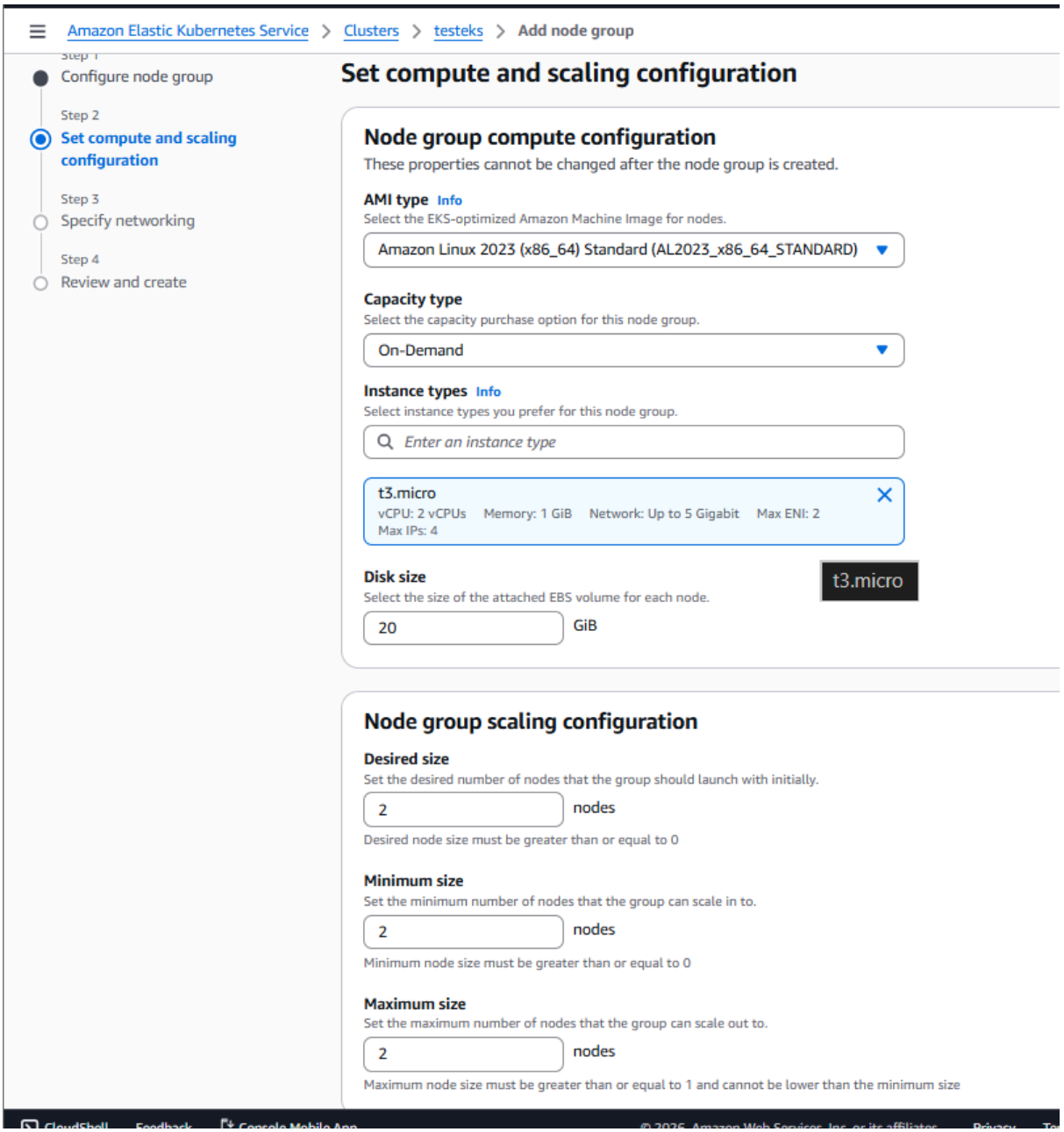
Create

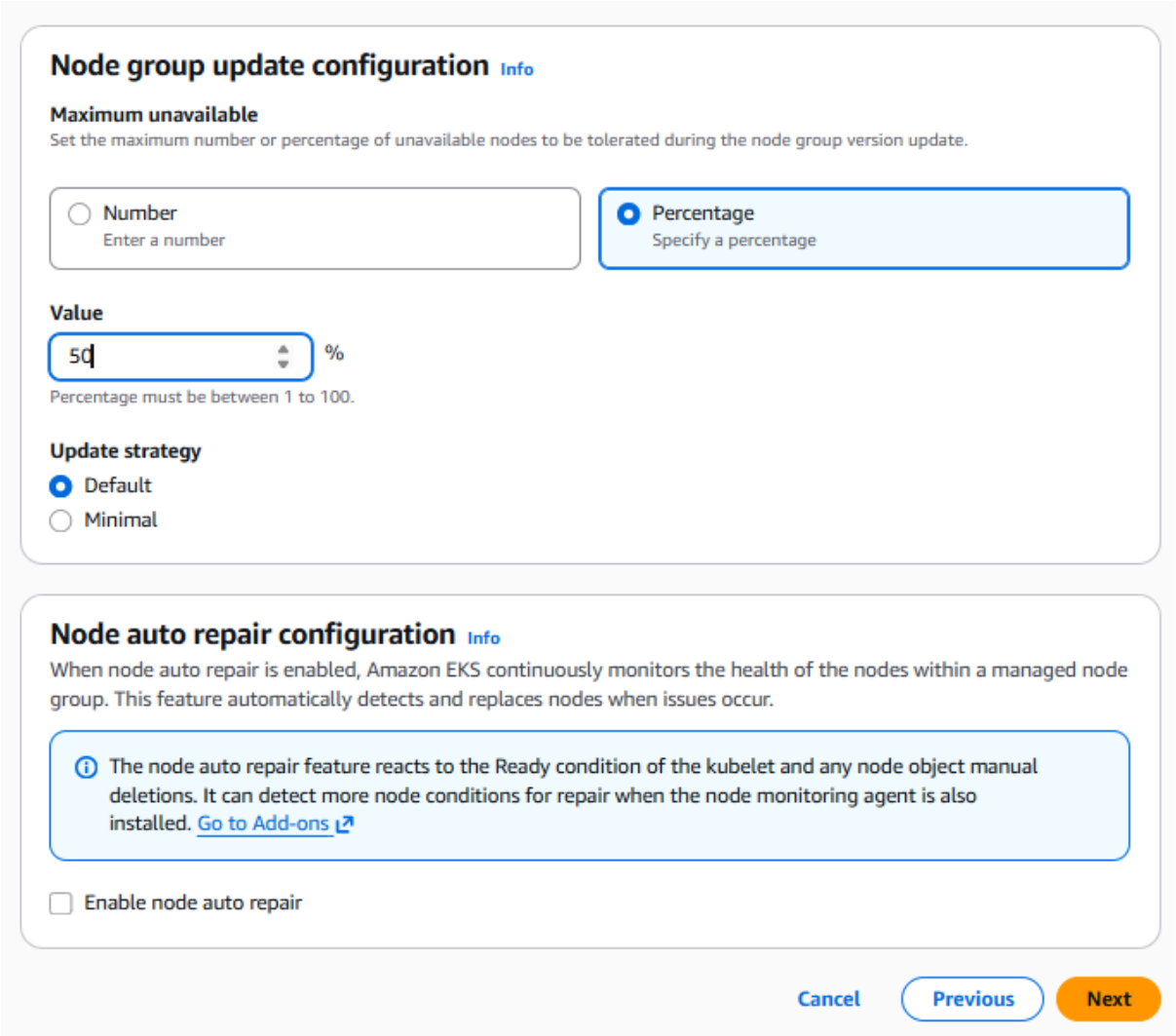


Created role



STEP 2:


Amazon Elastic Kubernetes Service > Clusters > testeks > Add node group
Configure node group
Step 2
Set compute and scaling configuration
Step 3
Specify networking
Step 4
Review and create
Set compute and scaling configuration
Node group compute configuration
These properties cannot be changed after the node group is created.
AMI type Info
Select the EKS-optimized Amazon Machine Image for nodes.
Amazon Linux 2023 (x86_64) Standard (AL2023_x86_64_STANDARD)
Capacity type
Select the capacity purchase option for this node group.
On-Demand
Instance types Info
Select instance types you prefer for this node group.
Enter an instance type
t3.micro
vCPU: 2 vCPUs Memory: 1 GiB Network: Up to 5 Gigabit Max ENI: 2
Max IPs: 4
t3.micro
Disk size
Select the size of the attached EBS volume for each node.
20
GiB
Node group scaling configuration
Desired size
Set the desired number of nodes that the group should launch with initially.
2
nodes
Desired node size must be greater than or equal to 0
Minimum size
Set the minimum number of nodes that the group can scale in to.
2
nodes
Minimum node size must be greater than or equal to 0
Maximum size
Set the maximum number of nodes that the group can scale out to.
2
nodes
Maximum node size must be greater than or equal to 1 and cannot be lower than the minimum size



Step 3 => default =>next
Step 4 => check => create

Now the node is created

kOps
https://kops.sigs.k8s.io/install/
website to install kOps

Cli commands

```
curl -Lo kubectl https://dl.k8s.io/release/$(curl -s -L https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl
chmod +x ./kubectl
sudo mv ./kubectl /usr/local/bin/kubectl
```

```
[root@ip-172-31-13-182 ~]# aws eks update-kubeconfig --name testeks
Added new context arn:aws:eks:us-east-1:988176744101:cluster/testeks to /root/.kube/config
[root@ip-172-31-13-182 ~]# kubectl get nodes
NAME                              STATUS   ROLES    AGE   VERSION
ip-172-31-69-67.ec2.internal      Ready    <none>   70m   v1.35.2-eks-f69f56f
ip-172-31-90-130.ec2.internal     Ready    <none>   70m   v1.35.2-eks-f69f56f
[root@ip-172-31-13-182 ~]#
```